

HCI REPORT – WILDCALL

Project Name: WildCall

Developer: Francesca Gaeta

Value Proposition: A bridge between People and the Planet.

The application is designed to provide community safety, ecological research and personal enrichment. The platform empowers users to interact with the environment responsibly and knowledgeably.

1. **SAFETY FIRST:** the app functions as a real-time radar, prioritizing not only human safety, but also animals' safety. We enable tracking of potentially dangerous wildlife sightings in a community-driven approach. Conversely, the platform maps common, harmless animal habitats (e.g., deer, birds, squirrels). This transforms trips into memorable adventures, offering users and especially children the chance to experience close encounters with local fauna. It is particularly useful for people who travel a lot to totally different countries or even continents and don't know everything about the local wildlife.
2. **ECOLOGICAL INSIGHTS:** every user is transformed into a citizen scientist since the collected data is invaluable to scientists and experts, providing detailed information on animal movements, including migrations towards urban environments, which is a growing global concern. By monitoring species location and behavior patterns, the application can offer useful data points regarding population health and species at risk of extinction. The platform facilitates free donations to verified green organizations and researchers, providing a direct channel for users to support conservation efforts financially.
3. **EDUCATION:** users gain access to expert-linked articles, detailed animal descriptions, and interactive games designed to educate the public on biodiversity, and critical environmental topics.
4. **DATA MONETIZATION:** embedded geographical maps allow us to passively collect aggregate, anonymized data on walkable paths, user habits, and general useful statistics. This structured information represents a further monetization opportunity for developers.

The application adopts a fully responsive design architecture to maximize its utility across all contexts. The content is primarily optimized in a mobile-first approach since it seems the most probable way in which users will dynamically interact with the app, but it is also designed for usage on larger tablet and desktop screens, especially concerning the learning part where people might be comfortably sitting at home, wanting to acquire new knowledge or killing time with instructive games.

CORE PROBLEM -> the dangerous and inefficient intersection of human activity and wildlife, taking into consideration also the unpredictability of some animals' behaviors due to environmental changes.

OUR SOLUTION -> a citizen-science application issuing safety alerts for dangerous wildlife crowdsourcing vital data to empower scientists in tracking phenomena like urban migration and species health.

NEEDFINDING

THE DOMAIN: ENVIRONMENTAL SUSTAINABILITY

The application was born to be introduced into the huge domain of environmental sustainability, which refers to using resources in ways that meet present needs without compromising the ability of future generations to meet theirs. Our focus within this domain is specific and actionable: the protection of local ecosystems and biodiversity, with a particular emphasis on animal welfare and conservation through data-driven insight.

This application was conceived as a strategic instrument that wants to go beyond typical user engagement. While it is presented as a playful, accessible tool suitable for any outdoor enthusiast or curious traveler, it inherently hides a deeper investment. The true purpose is to serve high-quality data and solutions for major environmental concerns, empowering researchers and experts while simultaneously engaging the public at any level of expertise.

Unintentionally, our core domain is merged with another theme that is the area of *Playful Exploration of the world*. This is due to the fact that, by providing a service to help users discover new paths and experience nature during travels, even during the ones that are not that far from their homes, we are automatically focusing on a segment that is exploring some places of the world. The playful layer conveyed by the gamification part is another driver of motivation for users, supplying the application's most profound function: contributing to real-world ecological research and fostering a global shift toward practices that minimize harm to the planet.

This theme immediately captured my attention due to a strong personal affinity for environmental topics. I often found myself conducting research and projects concerning the green sphere mainly because of school activities, and this academic background made me confident about such topics and taking the chance to leverage digital transformation to effect positive change. This level of personal commitment ensured the development of a product that is both professionally rigorous and personally engaging.

INTERVIEWS

The interviews were mainly conducted online (mainly texts) inserting an initial disclaimer to explain the scope and purposes of the project but, when possible, I wanted to collect information from users in person. This is how I managed to perform a kind of interview in which I did my best to remain as neutral as possible, without providing any suggestions to the interviewed, but reserving myself the right to intervene with further sub-questions in case the interlocutor came up with an interesting interpretation or idea, which was a less feasible way of interaction via cellphone.

I selected ten people to respect the principle for which we do not need a huge cluster as in statistics analyses to be able to extract the right insights to ideate and prototype a human-computer interaction project, and I purposely chose people that I knew were travelers and nature lovers, which made them more likely to bring useful observations. I paid attention to target as much as possible the types of users useful in interface development processes:

- **IMMEDIATE USER:** the user that will likely use the product, the reason why I chose people interested in such topics
- **EXPERT USER:** unfortunately, I did not have the chance to find some real expert of the domain. I tried to grasp the knowledge I needed from online material, videos, and examples of some similar applications reviewed by experts.

- **LEAD USER:** to respect this parameter, I thought of including in the interview stage people from totally different countries that I personally know, or I asked for help to colleagues currently on Erasmus to send the questions to international people that I considered useful in this context and that experience the problem of bumping into animals in the proximity of urban areas with a higher frequency with respect to our reality in Italy (unless we consider maybe mountain areas). In this way, I was able of acquiring insights also from Australia, North America, Mexico, Finland, some Asian countries. Including in my sample acquaintances that have often gone on big travels exploring places far away has also helped a lot. Some of them were already aware of some other similar applications already existing and some of them even used them, which made them a good target.
- **EXTREME USER:** I included some people that are passionate about travels, go into the mountains a lot and have always found themselves in solo travels or in more extreme situations, like going for trails or climbing mountains, which put them at a higher risk with respect to more moderate users that may not always go into the real wild or do some perilous things alone. As a matter of fact, they helped me understand some other attentive features that made me realize their high confidence with the domain.

Gentle disclaimer put in the questionnaire as if we were talking on person after explaining the University project to let the interviewed understand the domain and main topics:

<< In the following questionnaire composed of open questions (you can be as talkative as you want and provide as many examples and information as you want), I will ask you some questions that will help me understand what needs users may have in this context to check if I have correctly thought of all possible features able to satisfy them, if something original that I had not yet considered pops out, and this is all done to improve the app and make it user-friendly, intuitible and valuable.

I am sorry if this will take a bit of your time, but I wanted to let you know that your help would be great, and I would really appreciate it. Thanks a lot! >>

Afterwards, I split the structure of the questionnaire into thematic subcategories that I exploited to guide the crucial considerations I wanted to extrapolate.

DEMOGRAPHIC SECTION:

1. How old are you?
2. Where do you live (mostly)?
3. Do you consider yourself passionate about animals and general environmental themes?

BEHAVIORS AND NEEDS:

4. Do you travel often?
5. What kind of travels do you generally organize?
6. What activities do you usually do outdoors?
7. During your activities and travels, has it ever happened to you to encounter wild animals?

INTEREST IN THE APP:

8. Do you think you would find a similar app useful to your needs? If so, how?
9. Would you use this app only in relation to your outdoors activities or would you enjoy exploiting it in your daily life, too, to get some knowledge about a topic that might be interesting to you?
10. Can you think of any functionality and/or way of learning new stuff that you would expect from the app?
11. Which categories of people do you think may benefit from a similar application and why?

PAINS AND CURRENT PROBLEMS:

12. Do you currently face any difficulty when you try to collect data about some local biodiversity? If you can, some examples of your personal experience are very welcome

13. Are you afraid of getting into dangerous animals on nature trails or has it already happened to you?
14. And are you afraid that you may happen to encounter dangerous animals (or simply wild animals) in urban centers close to you or has it already happened to you?
15. Is there something that particularly worries you about animals' migration to cities and urban areas?

INTEREST IN ADVANCED FEATURES:

16. The app presents many free functionalities, but would you be interested in paying to access advanced features knowing that part of the proceeds would be devolved to researchers and organizations concerned about this topic to foster environmental protection?

INTEREST IN THE RESEARCH COMMUNITY:

17. Do you think such app could be useful to researchers and even citizen scientists that may not be forcedly experts but just enthusiasts that contribute to research? If so, how?
18. Would you like to actively contribute with observations and data collected during your activities or do you think you would use the app mainly passively to organize your itineraries and to acquire new general knowledge thanks to data from the community?
19. Do you deem it to be helpful to mix experts and normal users' contribution in a single app? If so, why?

CONCLUSIONS:

20. You came to the end! Just a final general question to collect a wider perspective eventually. Would you add some considerations or recommendations that have not been quoted in the questions above?

GENERAL RESULTS

Many users consider themselves passionate about the domain of inquiry. Holidays and trips are organized with family or friends, sometimes alone, especially for something less demanding, and they generally organize fun and adventurous activities. Utterly relaxing trips are less frequent; they like to stay dynamic. Many have encountered animals (most times not particularly dangerous) and they support the idea of having an application on their phone to check where these animals can be found in every different area they visit. Some of them state they would use the app also in their daily life to enrich their personal culture, and some have proposed games among learning methods, which was extremely positive feedback to me since it was something I wanted to implement, and I was trying to discover the existence of this need through some types of questions.

From the moment I wanted to attempt to include a monetization system that went beyond spatial data, I noticed that some people would actually pay for additional features, but in this specific case in which the answers are not totally anonymous we should remember that maybe the users felt somehow forced to say it since it was bound to charity as well. This is the reason why, thanks to insightful answers from an extreme user, I had the opportunity to revisit this concept and switching from a freemium base – where there were different packages at different costs to decide the kinds of functionalities and the duration of the subscription – to a still freemium-driven approach, but with a different purchase configuration that I will explain in due time.

They find the app both actively – signaling animals, playing games, interactions - and passively – checking uploaded sightings and reading articles – useful, and they feel like it is useful to mix up researchers and non-experts to dually contribute to the cause. Of course, some highlighted the importance of accessing reliable information in the app, hence the usefulness of efficiently dividing functionalities and parts of the application exclusively for experts and the others for normal citizens, which is something that I already intended to do.

INEDED IDEAS

As previously hinted, the interview was useful to reconsider some development scenarios. The initial idea was to provide three different paid packages, each with a name that recalled something natural. In order, Root (base plan, \$7), Oak (regular adventurers, \$35) and Forest (ultimate experience for dedicated naturalists, \$55) and the key discrimination was time-based, meaning that the activation for the first plan would have lasted 1 month, the second one six months and the latter one year. This distinction was made because, since the main driver was charity, I imagined users would have chosen to send their contribution to organizations and researchers depending on how much time they would have needed the application (ex. a student leaving for Erasmus or Overseas facing a totally different country maybe just wanted to activate the service for the semester or the year abroad because it would have been useful in that moment).

I immediately realized, anyway, that it was becoming chaotic and that it was not really hitting the main point, so I needed to find something better. I was lucky enough to interview a decisive colleague in presence, so when he told me that he would find useful in my contests some functionalities taken from a similar application he used to organize his trail runs, I could push the conversation further and I tried to take inspiration from it. After this conversation, I created a new mechanism based on downloadable maps. The charity principle is still there, but a percentage of the gains will be withdrawn by the creators of the app, and the remaining will be devolved to inherent associations because I value it as important and added value, but now the plans are organized such that:

- **PUPPY** represents the vanilla free plan, including functionalities like creating paths and exploiting the navigator mode, they can discover whatever information concerning the nation they live in that is known thanks to the "Country" field in the sign up page, but for any other place outside their country they have the possibility to buy the specific map divided per region, possibility to report animals and see the chronology of what others have signaled (but only for maps they have access to or that they bought), alerts if some problems occurred on the street (ex. temporary prohibited route for flood)
- **LION** is the paid version (£36/year, therefore \$3/month) that includes new features such as receiving notifications in real-time if someone reports any encounter very close to (or on the same way) the path you are walking or that you have saved and want to follow soon, no adv, accessing offline maps, no need to buy specific maps since everything is included

The marketing idea behind it is that people who travel a lot and know that they will make the most out of this application will automatically opt for the paid plan, gaining constant cash flow, while more occasional travelers (especially outside their belonging boundaries) that don't need additional functionalities will maybe decide to buy single maps on demand. Of course, they will cost more than a monthly installment, and the amount will also depend on the vastness of the selected region, but it will still be a saving for them if they need this solution occasionally. Plus, we could bring it to the next level adding a subtle gamification that will visually engage the users: they can see on their profile a planisphere where the activated maps become colorful, so that they can keep track of the places they have visited all over the world.

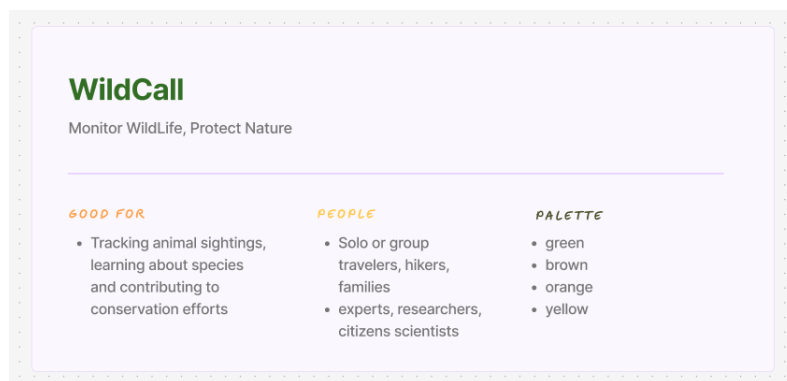
Another plausible user proposed to expand the same principles to plants, too. It would be surely coherent. Maybe this vegetal dimension could be implemented in the future and be released as a new version of the application. Additionally, some nice considerations came from another user saying that a worrying fact to her was the risk of hitting animals with the car running on non-strictly urban streets since it is very common to find deer and similar in her parts of America, and she also thinks that the application can be used in schools for learning purposes, especially with children guided by their teacher.

Just a single person in the whole sample was less enthusiastic about the application, saying that information can be easily found on the Internet nowadays, that she likely would not use the application because it is not something linked to her habits and needs, and this is totally normal since we can't conquest each single consumer on the market. Anyhow, she recognized the usefulness of the service for people that were more into this stuff, providing general positive feedback at the end.

The deepest user needs that I find valuable to my domain that I wanted to highlight are:

- 1) Risk mitigation for those who frequently engage in outdoor/extreme activities. Users are afraid of getting into dangerous animals on nature trails or encountering them near urban centers. Users want an application on their phone to check where these animals can be found in every different area they visit. The LION plan's core feature is receiving notifications in real-time if someone reports an encounter close to their path, but still in the PUPPY plan they have access to the chronology of what others have reported.
- 2) Reliable, organized biodiversity knowledge for itinerary planning. Users travel often and organize fun and adventurous activities outdoors. They want to organize their itineraries, but they face difficulty when they try to collect data about some local biodiversity. Users highlighted the importance of accessing reliable information and the need to efficiently divide functionalities for experts and normal citizens.
- 3) Active contribution and community-driven research. Users want to actively contribute with observations and data collected. They believe the app could be useful to researchers and even citizen scientists.

BRAINSTORMING AND PROJECTION USING FIGMA



COME LEGGERE IL BRAINSTORMING

L'intervista viene divisa in aree tematiche per porre l'accento su cosa si è indagato per ricavare certi bisogni degli user

Nella descrizione e progettazione dell'architettura dell'interfaccia, ogni titolo (ex. Home, Gamification...) corrisponde a un'unica schermata da sviluppare e visualizzare sul dispositivo

Le frecce portano all'approfondimento di una feature

L'intervista appare in inglese perché è stata mandata a possibili user in varie parti del mondo per guadagnare una visione più ampia del contesto che può essere molto diverso a seconda del continente, della zona in cui si vive etc.

Serve a capire le potenzialità di utilizzo a livello mondiale dell'applicazione, in quali mercati potrebbe riscuotere maggior successo, e ascoltando esperienze variegata si possono cogliere particolari e bisogni che non avremmo immaginato se abituati ad un ambiente diverso e relativamente più tranquillo.

LOGIN/SIGNUP

SIGNUP:

- email
- country (ovvero la mappa di colore di default alla zona)
- chiedere la lingua (si può sempre cambiare nelle impostazioni)
- username
- password
- ripetere password

Protezione Dati

RICONOSCIMENTO AL SIGNUP:

- habitat scelto → step in successione con nome top (la possibilità di skip)
- chiedere autorizzazione ad usare GPS (se non attivo anche quando si è fuori dall'app, questa funziona in background a manda notifiche di avvistamenti)
- autorizzazione ad altre notifiche push-up

Protezione Dati

LOGIN:

- username
- password
- Stato "ho dimenticato la password" (a risposta per mail)

Protezione Dati

• username o password sbagliati

- segnalare account inesistente se mail non nel db al recupero password

Protezione Dati

LOGIN RICERCATORI:

creare un'icona apposita per ricercatori registrati di nome/questi in cui bisogna solo mettere un ID e password e si accede a funzionalità esperti

Protezione Dati

HOME

TASTI di collegamento ad altre pagine dell'app:

- segnalare un avvistamento
- gioco
- controllare avvistamenti in determinate zone
- articoli

Protezione Dati

FOOTER

- disclaimer su informazioni verificate
- link social

Protezione Dati

GUIDA ALLE SPECIE:

separazione a società:

- specie marine → marino, acqua dolce
- specie terrestri → mammiferi, uccelli, insetti e anfibi
- specie arboree
- specie volanti
- specie artiche e polari
- specie arboree

connessione a DB con deviazione di caratteristiche principali habitat etc.)

Protezione Dati

RECENT SIGHTINGS:

recente

Protezione Dati

MAPPE DA SBOCCARE:

per chi ha il free plan, altrimenti alternativa PUPPY PREMIUM

- PUPPY free plan (creare percorsi, mappa del proprio paese sboccata automaticamente, le altre da acquistare, navigazione, avvisi se ci sono problemi dei percorsi salvati, tutto di challenge e a cui si può partecipare...), segnalato come piano attuale
- LUNA 360° (tutte le mappe anche offline, notifiche in real-time, via link di challenge...)

Protezione Dati

teme e decorazioni generali

Protezione Dati

SEGNALA UN AVVISTAMENTO

DEFAULT (si conosce la specie):

- Nome della specie
- Luogo dell'avvistamento
- Tipo di percorso urbano, naturalistico, marino, montano, non tracciato, ambiente estremo
- Nome descrizione dell'avvistamento, numero di animali, condizioni dell'habitat...

Protezione Dati

NON CONOSCE LA SPECIE:

- Riconoscimento con foto → carica foto (distinzione a tendina "la mia foto" o "foto da gallery")
- Riconoscimento con audio → carica registrazione (distinzione a tendina "registra ora" o "scegli da archivio")

Protezione Dati

GAMIFICATION

WILD QUEST:

- il database fa apparire immagini di specie a cui bisogna selezionare sulla mappa (in incrementi habitat naturale)
- dopo che si è inviato il tentativo, insieme al risultato esce una segnalazione nome dell'animale e dell'habitat

Protezione Dati

WILD QUIZ:

- domande tecniche o curiosità su animali e habitat
- si può filtrare per specie o luoghi

Protezione Dati

WILD CHALLENGE:

- 5 al mese con PUPPY, identico con piano premium LUNA
- più user che hanno scattato l'app possono collegare in una sfida di gruppo (gli avvisi si creano da loro) durante il loro viaggio o controllo alla scadenza chi ha vinto

Protezione Dati

CONTROLLA PERCORSO

collegamento con la mappa

Protezione Dati

si può tenere il nome di segnalazione e si vede la cronologia di avvistamenti

Protezione Dati

ARTICLES

elenco di articoli consultabili

Protezione Dati

AREA RICERCATORI (accesso riservato)

- continua la tua ricerca
- pubblica articolo

Protezione Dati

TASTO PER ANOMALIE:

quando un animale viene avvistato in un posto che non sarebbe il suo habitat naturale si mettono in coda anomalie

Protezione Dati

TASKS

SIMPLE TASKS: the most fundamental and high-frequency interactions. They represent the core, immediate value of the app and should be achieved rapidly and intuitively by a first-time user (from one to three steps, minimal decision making and data input).

TASK	PROCEDURE	RATIONALE
CHECK ANIMALS' LOCAL REPORTS	Click the Check button in the bottom bar and write the area you are interested in to verify whether an animal has been encountered along the path you want to walk	Test the core passive value of accessing community data (distinct scope for PUPPY and LION plans)
ACQUIRE KNOWLEDGE VIA ARTICLES	Go in the Articles page and find what they want to read	Tests the educational/cultural component of the app, which users mentioned they wanted for "enriching their personal culture."
ACCESS TO THE RESEARCHER AREA	In the log in page, if they are not normal users but researchers and have their credentials given by the organization they work for, they can just write their ID and password and will see a different version of the application.	Test the different features from normal citizens and experts, and the consequent capability of experts to supervise the material on the app to satisfy the truthfulness certification disclaimer.

MODERATE TASKS: require the user to interact with multiple features or complete a process that involves a sequence of decisions. It involves moderate cognitive load (e.g., configuring settings, comparing options, or navigating between different screens/tools).

TASK	PROCEDURE	RATIONALE
BUY A SINGLE MAP (free plan)	The Puppy user can do it by paying for the specific map with a variable cost depending on the size of the map. All information is found at the bottom of the Home page.	Test the alternate monetization strategy and the purchasing workflow for occasional travelers.
TEST OFFLINE MAPS	The Lion subscriber can freely download a new map and in case he loses connection will be able to use the navigator to safely move in the unknown environment and will still receive notifications.	Test the no connectivity features which are crucial for extreme users that may travel in isolated places that often lack connectivity.
REPORT A SIGHTING	Directly clicking the report sighting green button in the home page that redirects to the Report page (in the bottom bar), fill in the data (can be text, photos, sounds...), the headings clearly explain what the user	Test the primary active contribution and the clarity and ease of the input interface

	should do and the drop-down list helps simplifying choices	
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COMPLEX TASKS: are challenging, operations that may involve dependencies on external factors (like network connectivity). They check the app's power and its role as a platform.

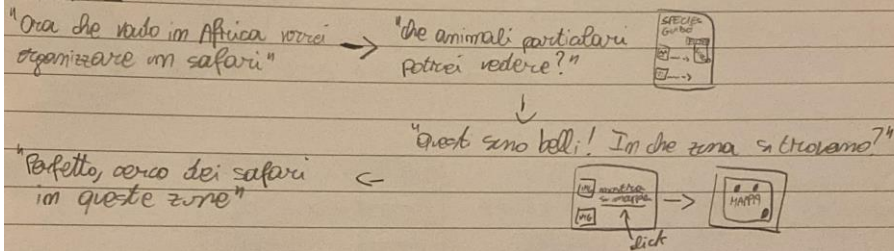
TASK	PROCEDURE	RATIONALE
START A WILD CHALLENGE	In the Games page, they scroll down until the Wild Challenge game and choose whether they want to take part in an already existing challenge (they can participate alone even with strangers) or if they want to create a group challenge with their friends. In this case, they personally write down their own assignment. Then they need to specify the day and hour in which the challenge will end, so that the ranking won't be updated anymore, and they can see who won. They can invite their friends by sharing a link from which they get directed to the challenge or via ID.	Tests the visual engagement and gamification element added to encourage exploration and continued use. It is probably one of the most complex tasks to set up in the application since it involves many clicks and internal/external connections.
ALL RESEARCHER FEATURES	The app is a sort of repository to them in which to save their research material and they can modify it in many ways (texts, images, comments, collaborations with others). It's a whole new platform to learn.	

STORYBOARD

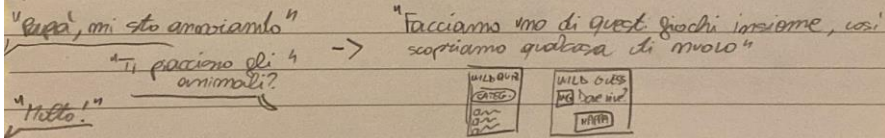
In the following, I attach the images related to the various storyboard scenarios I have considered because they are done on paper using primarily sketches and "comic book" approach because it is immediate to understand and easy to use. Thanks to the text and the speech bubbles, we have a better understanding of what is happening and the main idea intended to convey.

SKETCHES e STORYBOARDS

TASK 1

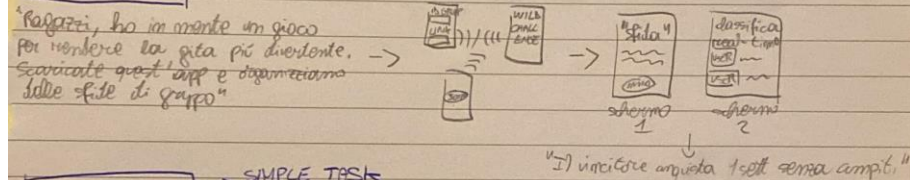


TASK 2



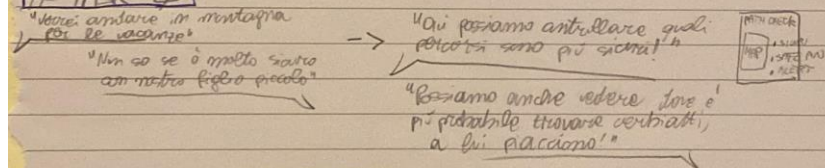
TASK 3

→ COMPLEX TASK



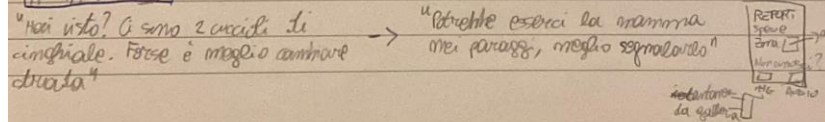
TASK 4

→ SIMPLE TASK

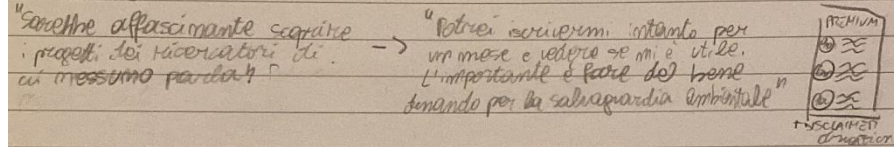


TASK 5

→ MODERATE TASK



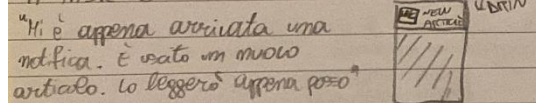
TASK 6



TASK 7



TASK 8



Storyboard → mix di "text-only" e "comic book"
 anche sketches

*Tasks 3-4-5 are the ones chosen during the A2 assignment.

The final storyboard I came up with is:



I think it is suitable to represent one of the main functionalities offered by the application, that is the possibility of checking where it is plausible to find some animals thanks to the map that a user might both buy on the fly with a “puppy” plan or having fully available with a “lion” plan (so the storyboard also underlines this distinction). In this case, it is for a pleasant purpose, that is organizing a trip with the aim of looking for a specific, innocent animal that people would like to see, but the same reasoning can be done in case of perilous animals. This will be implemented probably in the “Check” page of the app. It is not the only main functionality, but for sure is one of the most striking ones and, from my point of view, is the one that best envelopes the core project of wanting to track animals for different kinds of objectives.

LOW-FI PROTOTYPES

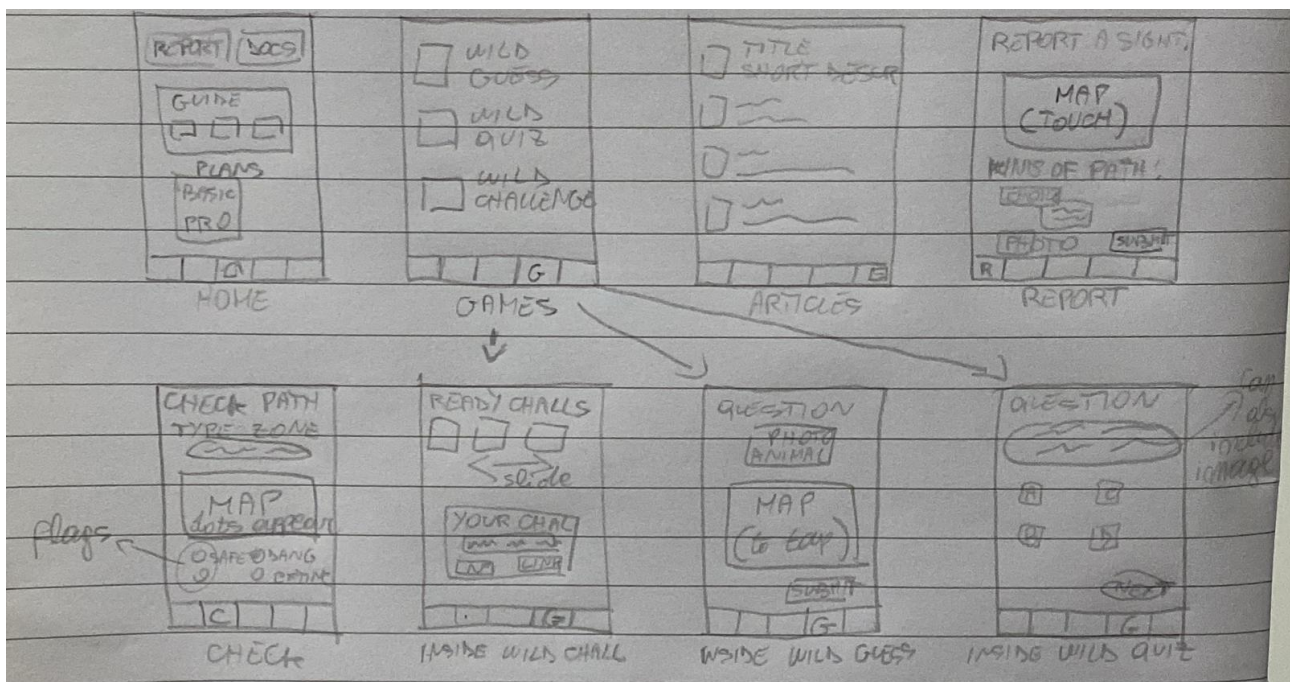
Given that the application helps users track animal sightings, stay safe, observe wildlife responsibly, plan routes, and access educational content, several interaction modalities were explored:

- **MAP-CENTRIC INTERACTION:** Users mainly interact with a large dynamic map that shows animal sightings, safe and unsafe zones, and preferred routes. It's intuitive for activities like sighting tracking and route planning, but can become cluttered and requires good spatial understanding from users.
- **LIST-BASED INTERACTION:** A list-first approach where sightings, articles, safety updates, and suggested paths appear in a vertical feed. It's easy to scan and scroll, works well for educational content, lower cognitive load for non-technical users, but less effective for spatial decisions.
- **GAME INTERACTION:** A playful interface where users earn points for reporting sightings, completing safety quizzes, or exploring animal profiles. It's very engaging, especially for

families with children, and encourages frequent use and learning, but might distract from safety information and could make the app less professional for researchers. Also, it could be too heavy and focused on gamification, while there is no need to focus the whole usage on games since we have already dedicated a whole page to it.

- **AUDIO INTERACTION:** Voice commands for logging sightings or receiving safety alerts (e.g., "Report a deer sighting", "Warn me if I approach a dangerous area"). It's useful when users are hiking and hands may be busy, but it's harder to prototype in low-fid and not reliable in noisy environments.
- **NOTIFICATION-DRIVEN INTERACTION:** Push alerts about dangerous areas, new sightings, or nearby species of interest. It can annoy if overused, not all users may enable notifications because not everyone wants them and the aim is letting them free to use the app when and as they want because the emphasis should be on getting information for one's own safety or curiosity e.g., when they need to organize a travel or while they are walking and encounter an animal, so a constant update could be useless for the main purposes of the app.

After evaluating the modalities, two primary approaches were selected for the low-fidelity prototype because they best support the project's goals: map-centric interaction because core tasks like sightings, route planning, and identifying safe/unsafe zones all rely on spatial awareness, and it's ideal for hikers, tourists, and researchers; and list-based approach for secondary screen because it's perfect for educational content such as articles, species descriptions, and gamified quizzes and it's easy to prototype. The two modalities also complement each other.



Because the app is intended for hikers, tourists, researchers, and families, accessibility and portability are essential. The main target platforms are mobile applications primarily with Android and iOS, and a responsive web version as secondary platform to plan routes and read articles before a trip. The main target device is smartphones because it will be useful for the main use cases during hikes, trips and wildlife encounters. Then it can be suitable for tablets and desktop browsers for people who are comfortably sitting at home and have time to really get informed and learn while reading articles and playing games and exploring maps on bigger screens.

The screens that I have prototyped consider smartphone devices because it's the most important use case. The drawings included should represent in the present moment all the interfaces for the different activities to offer to users. From the Home screen, thanks to the bar at the bottom, we can move to all different pages. There are also other links, like clicking on the "report" button brings to the page with the map to signal a sighting, and the "docs" instead bring to the list of documents in the related page. The Games page has been branched into three games that we can play.

MEDIUM FIDELITY PROTOTYPES



To create this second set of more reliable prototypes, I used FIGMA to convey a very tangible idea of what the application will probably look like. Of course, using a tool like FIGMA I had the chance to be very specific in certain details (like images taken from the Internet, longer pages scrolling down since I did not need to put effort into drawing by hand, etc.).

Some *violations* of basic design principles encountered are:

- the unavailability of an arrow/button to go back to a previous page or action

- some non-clear icon images that need to be more customized like the ones for the games
- pages for login/signup missing
- no possibility of accessing some settings, and consequently all the useful settings missing
- a real map is missing
- no real connection to the camera and audio to upload photos and recordings
- no connection to an embedded database to retrieve all sightings and updates from the community
- some development ideas are not respected in this medium-fidelity representation because they concern topics that came to my mind later, such as switching from three different plans to only two (the free and the paid ones) because the new idea was charging users with various price ranges depending on the map they needed to buy, because they can only see the territories where they live when at the subscription they have to select their country
- they can't manually select points on the maps to signal a wild animal on the Report page or to check for sightings in a specific area in the Check page, so this is a functionality to implement in the real app
- users should have the possibility to enable notifications for the information they want and even only temporarily, such as for receiving updates on new articles which were published, or for a recent sighting in the area where they live or for the areas they are planning to visit in a future trip or for something they have already searched

link to access med-fid prototype:

<https://www.figma.com/files/team/1556227741824988140/resources/community/file/1573430756236630726?fuid=1556227739489430826>

HIGH FIDELITY PROTOTYPE

- A. **TOOLS, FRAMEWORKS AND LIBRARIES:** The prototype is built using the following technology stack:

CATEGORY	TOOL/LIBRARY	REASON
Fronted Framework	React	Selected for its component-based architecture, which enables modular, reusable, and efficient UI development.
Language	TypeScript	Used to enforce static typing, significantly reducing runtime errors and improving code quality and maintainability.
Build Tool	Vite	Chosen for its speed in cold start times and hot module replacement (HMR), drastically improving the development experience compared to older bundlers.
Styling UI	Tailwind CSS and shadcn-ui	Tailwind (utility-first CSS) enables rapid styling. shadcn-ui (built on Radix UI) provides accessible, high-quality, pre-built components (e.g., Card, Button, Dialog) for a

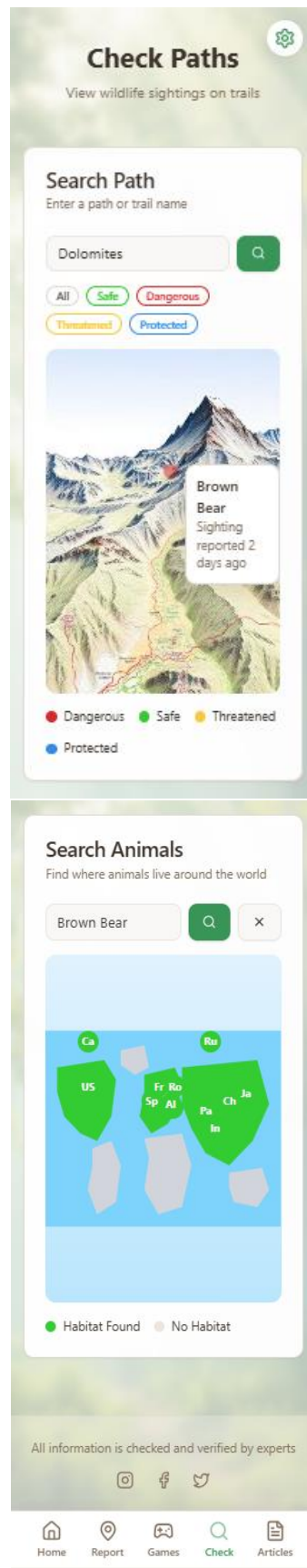
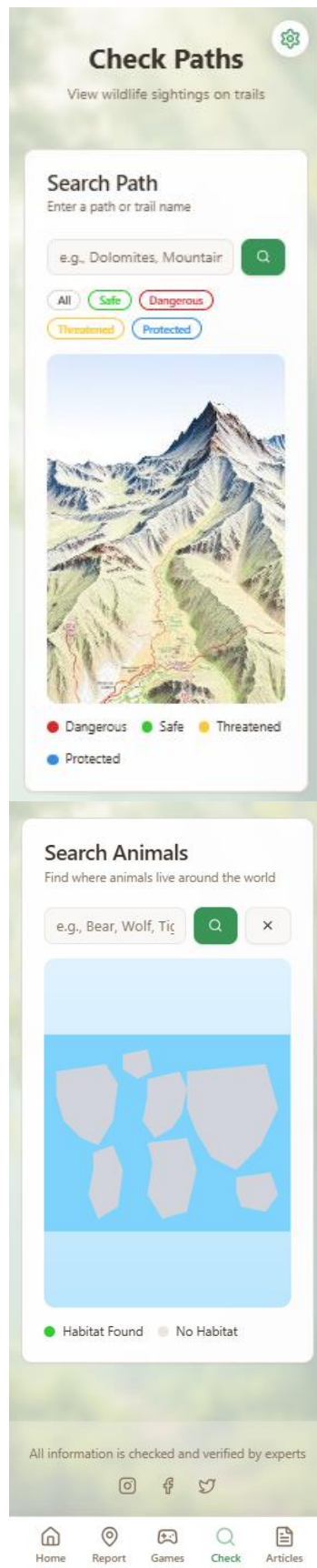
		professional, customizable look.
Routing	React Router DOM	The standard declarative library for navigating between different pages and views within the single-page application.
Backend (database)	Supabase	Provides a complete, "backend-as-a-service" solution, including PostgreSQL database, Authentication (Auth), and API access, minimizing manual backend development.
Data Fetching	React Query (TanStack)	Simplifies asynchronous state management by handling caching, background data fetching, and synchronization with the server.

B. **Git Repository Link:** <https://github.com/Francesca9933/WildCall>

C. **SIGNIFICAT DESIGN CHOICES:** a **nature-inspired color palette** (forest green, warm orange, brown, golden yellow). It is defined in *src/index.css*. This palette is complemented by fixed image backgrounds on core pages (*Home.tsx*, *Report.tsx* etc.). Interactions are themed with nature branding buttons. The ***Tutorial.tsx*** component is conditionally displayed on the *Home.tsx* page for new users. This prioritizes a clean first-time user experience (FTUE) by explaining the main features (Report, Games, Check) upon initial login, improving feature discoverability and retention. The use of ***BottomNav.tsx*** ensures the main navigation links (Home, Report, Games, Articles, Check) are always available and easily accessible with the thumb.

D. **SIGNIFICANT SCREENS OF THE PROTOTYPE:**

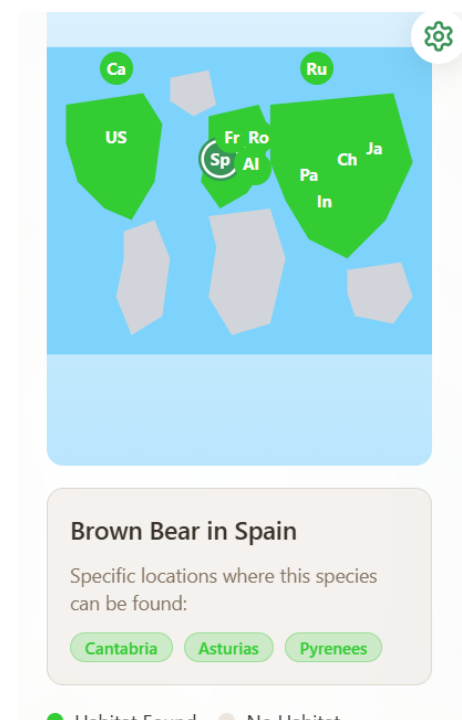
SIMPLE TASK #4: Checking for animals signaled on the path one is considering walking; or the user could decide to look for a list of places where a specific animal has been found, either to avoid it or having the chance to see it (e.g., for children).



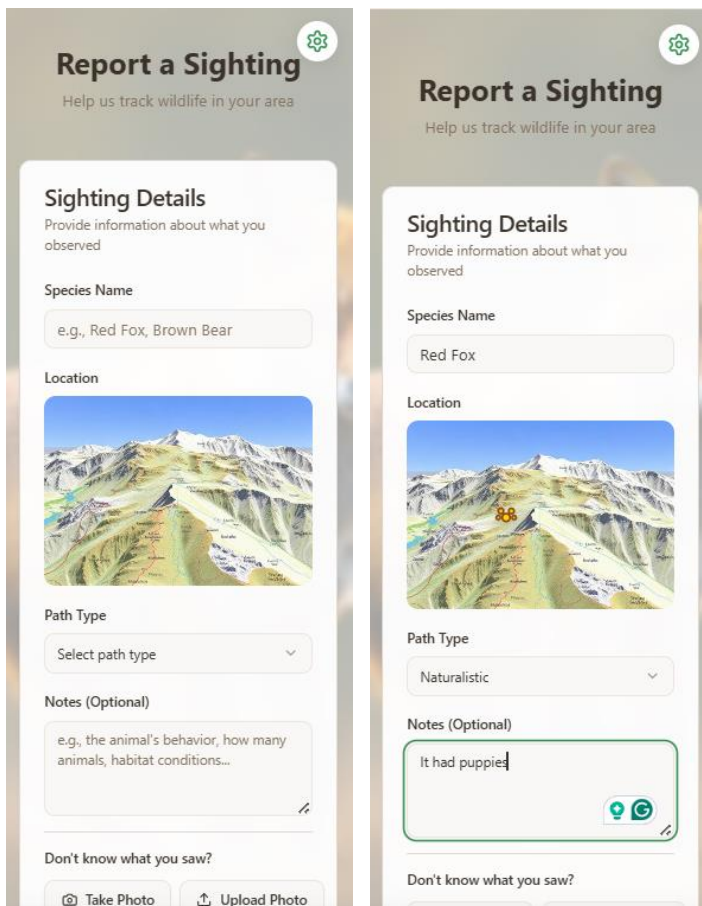
When we look for a generic area or specific path or road, clicking the search button triggers the appearance of dots that will have different colors depending on the level of dangerousness of the animal and other useful information in the flag.

When we type the name of an animal, a drop-down list of suggestions appear (ex. Brown Bear, Black Bear, Asian Black Bear while writing "Bear"). After selecting the animal, in the planisphere the places where you can find it become colorful (for simplicity I have mocked this unrealistic map because it was easier to implement the mechanism).

If we click on a specific country, more specific places appear:

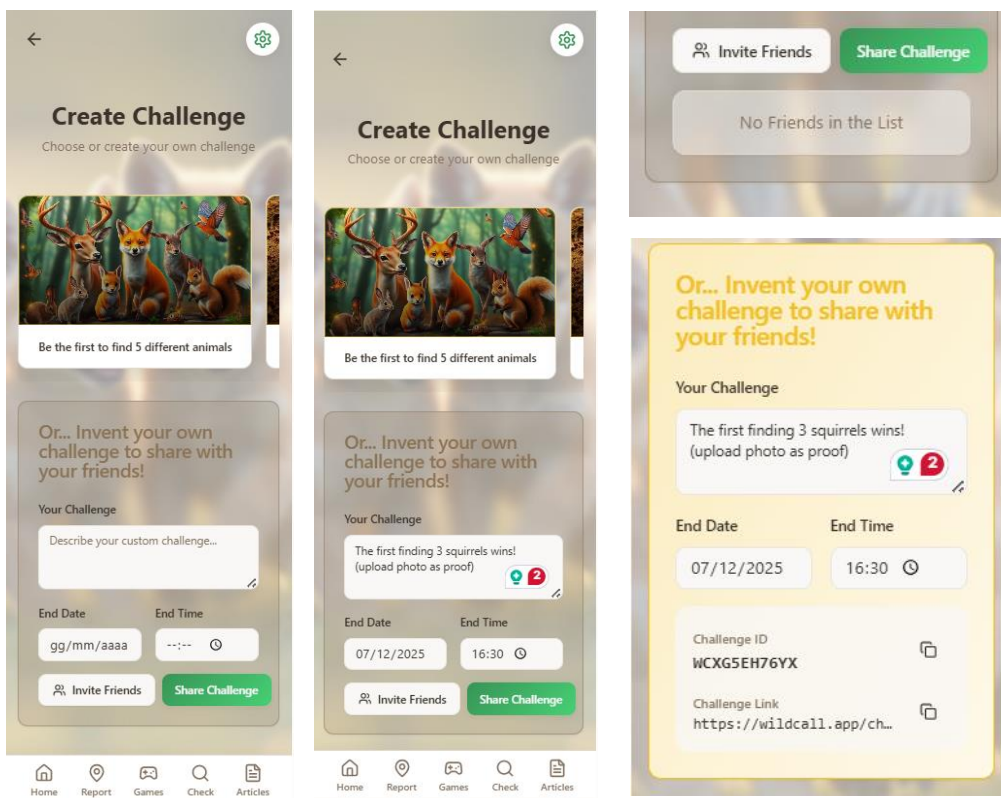


MODERATE TASK #5: Signaling the animal in the Report page.



We write the name of the animal (if we don't know it we can upload a picture or record its sound and the database will find a match). We tap on the map the place where we found it (we can also look for the place by writing it in the search bar, added in a later version after testing), we select the kind of path where we encountered it (Naturalistic, Urban, Marine, Mountain, Not Tracked, Extreme Environment) and this should be very useful to researchers. We can write additional notes.

COMPLEX TASK #3: Creating a Wild Challenge Room since it requires many different taps and parameters to configure, plus sharing link or code.



We can either choose one of the defaults challenges or create our own if we have more imagination or want to do specific activities. We write the text, set the end date and end time, and we can choose whether to invite friends directly from the friends list in the app (of course there is no friends registered now) or by sharing the challenge, which means generating the ID and the link to share via message.

Especially in the last set of screens it is visible the addition of the “back” arrow that was initially missing (in the other pages it is not present because they are independent and unique, so we don’t enter any further subfolder from which we might have to reverse the path). Furthermore, on the initial Games page I have personalized the icons to make them inherent to the general application’s mood. It is also clear that the “setting” button has been added.

In the final high-fidelity prototype, the login/signup pages have been developed, adding the targeted login interface for researchers with their official ID and given password (I did not implement also the part of the application that researchers should use because the code would have become immense for the purpose of this University project, plus it would make sense to really acquire some experts’ reviews to know how it could be correctly implemented. It may be some sort of update that will be released with future versions of the app).

Maps (AI-generated, not utterly realistic for simplification purposes) have been put where requested. The real connection to the audio is still mocked because, while trying to make it real, it kept on pointing to the phone camera and the gallery even though I specified that the button should direct to the voice memos and I can’t understand why this part of the code does not work anyways, but at least the real connection to the camera and the gallery correctly works.

Users can now manually select points on the maps. For instance, in the report page’s map they can tap on the map and a pointer shaped as a paw will appear and they can fill the requested fields. If they touch another part of the screen the previous paw will disappear because I used the assumption that if they re-tap it is probably because they got the previous tap wrong. Consequently, if they are not signaling the anima in real time, but they are waiting to go back home before submitting all the possible encounters that they made, it means that they can’t report more animals with a single compilation yet should iterate the process, but I think it is acceptable because animals are not spotted that often, so there is a very low percentage of seeing more animals in a single trip, and users can tolerate a little extra effort for those rare times when it happens. To respect one of the usability principles, if they click on an existing paw, it gets discarded, so that they have a solution to revert possible errors.

Finally, users can correctly enable push notifications directly from the settings, or also from the button in the article’s page that brings them directly to the settings (I did not develop additionally the specific settings’ voices internally for the same reason I did not explore also the specific interface reserved for researchers, it was becoming a huge code to handle with deadlines to respect).

- E. The application is designed as a mobile-first web application (The application utilizes a *BottomNav* component fixed to the bottom of the screen, a standard pattern for mobile applications to provide easy thumb access to core features), prioritizing a seamless experience while remaining functional on larger screens. The visual design is anchored in a "Nature-inspired" aesthetic, utilizing earthy tones and organic gradients defined in the global CSS and Tailwind configuration.

Responsive Breakpoints: A constant `MOBILE_BREAKPOINT` of 768px is defined in `use-mobile.tsx` to programmatically detect mobile vs. desktop environments, allowing the UI to adapt based on the user's screen width.

Touch-Friendly UI: The layout includes a fixed "Settings" button in the top-right corner, ensuring critical configuration options are always reachable. The main content is wrapped in a container with a maximum width of `screen-xl` and centered to maintain readability on wider desktop monitors.

Platform-Specific Polish: To prevent content from being hidden behind the fixed navigation bar on mobile, the main layout includes a `pb-16` (padding-bottom) class.

Nature-Inspired Palette: The color system is built using HSL values to represent natural elements (primary as forest green, secondary as warm orange, elements that should be more striking to the eye with golden yellow, the background follow beige tones using off-white to be less impactful).

Depth and Elevation: The system defines three levels of custom shadows (soft, medium, and strong) to create a sense of hierarchy and elevation between different UI components.

Typography and Shaping: A standard radius of 0.75rem (12px) is applied to cards and buttons to give the interface a modern, and rounded feel.

Motion and Interaction: Subtle animations such as fade-in, slide-up, and pulse-slow (used on active navigation icons) are implemented to provide feedback and improve the overall user experience.

F. COMMENTS ON CODE AND DATA

I inserted some hard-coded parts for UI elements that consequently can embody some limitations, but this has been done because since I have not connected nor developed a real, huge database with all the existing species of animals in the world or real sightings (because of course the app is not released and used by real users), they were essential to mock some important interactions for demonstration purposes.

Recent Sightings: The "Recent Sightings" list on the Home page is entirely hard-coded with specific examples (Mountain Goat, Bald Eagle, Gray Wolf) and fixed timeframes (e.g., "2 hours ago").

Search and Sighting Simulation: In the "Check Paths" feature, the logic is hard-coded to only trigger a result if the user types "dolomites." This displays a fixed notification for a "Brown Bear" sighting reported "2 days ago".

Subscription Plans: The pricing and feature lists for "Puppy - Free" and "Lion - Premium" are hard-coded directly into the Home page component.

Tutorial Triggers: The tutorial visibility depends on a specific metadata flag (show_tutorial) within the user's Supabase auth metadata, which is hard-coded to update to false upon completion.

The prototype uses complex local objects to simulate a database for the species search feature.

Animal Database: A comprehensive animalDatabase object in Check.tsx stores geographical information for several species (Bears, Wolves, Tigers, Lions, Elephants). This includes nested lists of countries and specific regions for each country.

Map Visualization: An animalNames array derived from the database keys is used for search suggestions, while a countryPositions object maps specific countries to approximate coordinates on a custom world map.

Species Categories: The SpeciesGuide and Report pages utilize pre-defined arrays for categories (Terrestrial, Aquatic, Urban, etc.) to populate the UI and dropdown menus.

As anticipated, as a prototype, several core features are currently non-functional or purely illustrative.

No Live Backend for Sightings: there is no dynamic "Global Sighting Feed." The report submission process in Report.tsx only triggers a success toast and clears the local form state; it does not persist data to a database for other users to see because it is not needed at this stage.

Map Placeholder Logic: the "Check Your Maps" and "Adventure Essentials" sections are static call-to-actions. There is no actual system for unblocking, downloading, or rendering real-world offline map data beyond a static image of a planisphere or a placeholder for simplicity and uselessness.

Simulated Media Capture: the "Take Photo" and "Record Audio" buttons currently only log the file name to the console and show a toast, rather than performing actual file processing or storage.

Static Articles: the Articles page contains three fixed placeholders (Conservation Today, Habitat Protection, Species Spotlight). There is no integration with an external news feed to provide fresh content.

Limited Navigation Logic: many navigation paths, such as "View Full Guide" or "Browse Maps," link to either placeholder pages or the top of the existing guide rather than a deeper, searchable directory.

USABILITY TESTING

To conduct an effective usability test for the *WildCall* application, I focused on the core user journeys: reporting wildlife, checking path safety, researching animals, having fun with the games and engaging with the community/premium features.

I asked **one of my colleagues** to be the user to test the application flow and simplicity. I opted for the **think-aloud** methodology because I wanted to listen to his reasoning to better understand what a user may notice in first place and how to improve the app. I also trusted him to try it because, since he is from my same master's degree, I knew he could have a better understanding of how an application should work and be evaluated, to provide me with some useful insights. I thought it was a good idea to mix "normal users" thoughts with a bit of competence to immediately extrapolate useful information, since I could not test on a wider range of people. Plus, he is keen on travel, trails in the mountains, he is a sportsperson and adventurous, and was interested in the app, therefore he represented the right target.

I impersonated both the **facilitator** following the **script** asking for permissions and tasks to be completed, and the **observer** since I was also the one taking notes on the user's behavior. The evaluation was conducted at my place. We were both in the same room because I did not have the equipment and the room at disposal to be fully professional as the normal procedure would require, but I kept silent outside the script unless he had some inherent questions. I let him access my application through the link provided by **Vercel** (where I deployed the app) to check the visualization also on a different screen from my iPhone, and I was sitting next to him while using the functionalities. I only explained the general application to the user, asked if he had further questions in the right moments, provided him with all the information before starting so that he could give his consent consciously, I specified to him that he was not forced to do anything and he could retract from the activity at any moment despite the signed permission, I remained impartial and only read him the text of the tasks that I tried to keep simple, realistic and useful for him as a user so that he could engage with the interface and did not leak suggestions, recalling him to think out-loud.

I chose to let him fill both the **SEQ** (post-task questionnaire with Likert scale from 1 to 7) because I wanted him to evaluate the task when the experience was still fresh in his mind and gain useful insights on how the task felt during a first interaction with the system and knowing which specific tasks required a better development, and the **SUS** (post-test questionnaire with Likert scale from 1 to 5 and consequent score calculation studied in class) at the end of the whole procedure, first of all because using the SUS alone would not maximize the result since we would grasp a general performance of the system but not what can be improved specifically, and secondly because the user had the right knowledge, in my opinion, to evaluate objectively and usefully the system, knowing what are useful characteristics to look for. I did not use also the NASA TLX because it is more suitable for complex environments like military, healthcare...

THE PERFORMED TASKS:

TASK 1: wildlife reporting

GOAL: Evaluate the intuitiveness of the "Report Sighting" form and the map-marking interaction.

CONTEXT: you are on a hike and have just spotted a wild animal

REQUIREMENT FOR THE USER: "While exploring a local trail, you've just spotted a Red Fox.

Use the app to record this observation so other hikers are aware of it. Include where you saw it and any other details you think are important"

TASK 2: safety and path verification

GOAL: Test the "Check" feature, specifically the ability to interpret the safety legend and search for specific locations.

CONTEXT: you are planning a trip and want to ensure your chosen route is safe.

REQUIREMENT FOR THE USER: "You are planning a weekend trip to the Dolomites. Before you go, check the app to see if there have been any recent dangerous wildlife sightings reported in that area that might affect your safety"

TASK 3: educational research

GOAL: assess the navigation of the "Search Animals" map feature.

CONTEXT: You are curious about where a specific animal lives.

REQUIREMENT FOR THE USER: "You've always been fascinated by Brown Bears. Use the app to find out which regions of Italy they are typically found in"

TASK 4: engagement and personalization

GOAL: Verify the ease of finding the settings, enabling notifications, and navigating to the educational articles.

CONTEXT: You want to stay updated on wildlife conservation news.

REQUIREMENT FOR THE USER: "You want to make sure you don't miss any new conservation news. Find where you can read recent articles and update your account so that the app alerts you whenever a new article is published"

TASK 5: interactive learning

GOAL: Test the "Games" section and the feedback loop of the quiz interface.

CONTEXT: You want to test your wildlife knowledge.

REQUIREMENT FOR THE USER: "You want to practice identifying animal tracks before your next hike. Find a way to test your knowledge with a quiz and complete the first question"

TASK 6: evaluating value proposition (premium)

GOAL: Determine if the user can distinguish between free and premium features and find the upgrade path.

CONTEXT: You are planning a long expedition where you won't have internet access.

REQUIREMENT TEXT FOR USER: "You are going on an expedition to a remote area with no cell service. Explore the app to find a way to access maps while offline and determine what steps you need to take to get that feature"

TASK 7: test a more complex task with more steps like creating a "wild challenge"

GOAL: Determine how feasible it is for the user to create a challenge and invite his friends and how easily he can interact with this more complex interface

CONTEXT: You want to activate a challenge with your friends

REQUIREMENT TEXT FOR USER: "You are traveling with your friends, and you want to customize your own challenge and invite them to play together during the trip. You want to see who will be the one to see more squirrels. Find the way to create the challenge and invite them"

TASK 8: unknown wildlife reporting

GOAL: Evaluate the intuitiveness of the "Report Sighting" form and the map-marking interaction for reporting an unknown animal.

CONTEXT: you are on a hike and have just spotted a wild animal that you don't know

REQUIREMENT TEXT FOR THE USER: "While exploring a local trail, you've just spotted an animal that you could not identify, but you succeeded in taking a photo of it. Use the app to record this observation by uploading the picture so that the database can associate the subject in your photo with something known in the database thanks to AI, so that you can both learn something and make the other hikers aware. In this case, there is no need to fill the other fields since it was already required in another task if you don't want to waste time"

In the attached PDF I supply the signed recording consent by the user and the usability test script followed along the process. In such PDF, at the bottom, you will also find the SEQ and SUS results, extensively explained in the following.

[See Usability Test Documentation](#)

Most of the tasks were extremely easy to fulfill, the flow was natural and all of them were correctly taken to the end without any aid. During some of them, some useful indications emerged, so for brevity reasons I will point out only the relevant observations that lead to an improvement of the current interface:

- ❖ During the second task, the user at the beginning considered the wrong part of the application. He thought he should have used the bottom part in the Home page called “Check your Maps” because of the misleading word “Check”. Then, reading the subtitle, he understood alone that it was not the correct path and immediately spotted the correct “Check” button in the navigation bar below, and the rest of the task followed naturally. Since the user valued this feature as one of the main functionalities coming with the app (together with “Report”), he suggested to change the initial orange button in the Home stating “Read Articles” with “Check Path” so that it could be more visible to the eye. It was easy for him finding also the alternative path from the navigation bar.
- ❖ During the third task, he was not able to find the Italian region with the mobile device in portrait mode, while it was visible in landscape mode because the map has more space to expand. This was the reason for a lower score despite the easy task to complete, but I chose to not consider it a major problem to adjust because I already hinted that this map was intentionally mocked for simplicity. In a real final version ready to be deployed it would be replaced by a real map with the right references to territories.
- ❖ The fourth task has the lowest score even though it could have been the easiest task because, since the voices in the settings did not furtherly land to an inner page, the notification enabling button in “Articles” brought to the general menu settings, and the user lost more time to read all the options and finding after two or three attempts the right voice for “Language and Notifications” (he first thought it could have been in “Account” or “Privacy”). Therefore, I decided to develop the code only for the Language and Notifications part (with minimalism) to create a direct link between the button in “Articles” and the right interface to make the process easier, which is the standard used also by all the other apps.
- ❖ In the sixth task, it was easy for him to spot the first solution of buying a specific map in the “Adventure Essentials” kit with the “Browse Map” button, also because he admitted he was attracted by the convenient starting price of 3 dollars, and did not consider opting for a premium plan. By rephrasing the task to have him to look for the most cost-effective solution depicting the user as someone who is always travelling in new places many times in a year, he noticed the “Lion” plan and said he would have compared the price of buying all the needed maps and see if it would have been more logical to directly purchase the premium plan.
- ❖ Task seven was actually very intuitive for him, but did not assign the highest score because he recognized that it took lots of clicks and could be a more complex task (as I had already individualized in previous sections of the report), so he took the length of the path for the task into consideration, too, but had nothing to complain about. He also noticed that it could have been less immediate to understand for a less digital savvy user (e.g., his mum) because it also implied sending links and code via different apps and it could be more demanding. This is also the reason why I took the chance to explain to him the usage of the Friends List that was not brought in surface during the test phase when he asked me about future plans for treating the app as a social media. During the usability test we tried out the most relevant tasks, but his genuine interest to me validates the idea of giving this social footprint.
- ❖ At the end of Task 8, the participant suggested making the option to add a picture or sound of an unknown species more immediately visible to users. Otherwise, if the user does not recognize the species, they may overlook this option and fail to report the animal. Based on this feedback, the order of the options was modified by placing the “Don’t know what you saw?” option at the beginning, thereby reducing potential misunderstandings.

I followed the instructions to calculate the SUS score with the points acquired. A SUS score of **87.5** reflects very high perceived usability. Users reported strong confidence when using the system, low perceived complexity, and minimal need for technical support. Such score suggests that the system is intuitive and accessible to a broad range of users. Something to specify about the ratings given to some questions of this second questionnaire is that he just evaluated the first question on a personal level, considering that he might need this app a lots when traveling, especially when he goes to the mountains during summer, but of course this is not his main activity, so he gave a mark in the middle not because he does not like it.

In the final part of the conversation, after the user's curiosity, I asked him how much he thought his previous knowledge and familiarity with similar applications might have helped him in understanding immediately the interface. He answered that his experience helps for sure, because he is used at using social networks and applications in general, but despite it he finds the whole system extremely easy to understand even for non-experts. He also highlighted the memorable and learnable trait of the application by saying that the initial scores at the conclusion of each task were stricter because he was just getting used to it, but after some navigation he felt already well integrated with the system, which made the other objectives easily reachable.

OTHER QUANTITATIVE METRICS:

Task Success Rate (Effectiveness): in binary terms, all tasks were completed. Specifically, all eight tasks were successfully completed without external aid, resulting in a 100% task completion rate. Minor hesitations were observed in Tasks 2, 3 and 6, but they did not prevent successful completion. Moreover, task 4 was completed, but with a bit more help in the second part about notifications activation, which is something that can affect other scores.

Time on Task (Efficiency): thanks to the recording, I can derive the exact time spent on each task:

- Task 1: 65sec (he was very specific on the descriptive information about the animal so the time dilatated, but not for complexity)
- Task 2: 48sec (with first error about clicking on "Check your Maps")
- Task 3: 125sec (2min5sec, more time because he found two possible alternatives and because we had initial problems with the mocked map as already described)
- Task 4: 100sec (problems with notification enabling)
- Task 5: 35sec
- Task 6: 15sec for the first solution about buying maps, plus 5sec rephrasing the task for cost-effectiveness, total amount 20sec
- Task 7: 60sec
- Task 8: 25sec

MEAN TIME ON TASK: $\frac{65+48+125+100+35+20+60+25}{8} = 59.75 \text{ sec}$

Time alone would not be enough, since speed needs to be combined with success, so the mean should be calculated considering at denominator only successful task, but in my calculus this is already true.

Learnability Using Time Reduction:

- Avg Time over first 4 tasks: 84,5 sec
- Avg Time over last 4 tasks: 35 sec
- Learnability Gain: $\frac{84,5-35}{84,5} = 59\%$ percentage of decreased time for task completion indicating strong learnability

Error Rate: number of wrong paths, misclicks, or misunderstandings. If a task involves these kinds of mistakes, we add a penalty that is a fixed increased amount of time to complete the task based on the seriousness of the error. Table of commonly used penalties in HCI:

Minor navigation errors/Brief hesitation/self-corrected	5sec
Wrong section/backtracking/extra reading	10sec
Major misunderstanding/task breakdown	15-20sec

Penalties should also be proportionate to the mean time for task completion, because they should be noticeable but not dominant.

Formula: Total Time for the Task + (Number of Errors * Penalty)

Task 2: $48 + (1 \cdot 10) = 58\text{sec}$

Task 3: $125\text{sec} + (1 \cdot 5 + 1 \cdot 20) = 150\text{se}$

Task 4: $100 + (3 \cdot 15) = 145\text{sec}$

Task 6: $20 + (1 \cdot 5) = 25\text{sec}$

Task 8: $25 + (1 \cdot 5) = 30\text{sec}$

Task	Raw Time (sec)	Errors	Adjusted Time (sec)	Increase
2	48	1	58	+21%
3	125	2	150	+20%
4	100	3	145	+45%
6	20	1	25	+25%
8	25	1	30	+20%

CONCLUSIONS

I truly enjoyed this course because I have always been passionate about creating websites and applications, and I appreciated the opportunity to develop a project connected to a topic I deeply care about. Working on an application focused on animal safety and monitoring made the experience especially meaningful, as it felt like something that could realistically be published on an App Store, be useful for real users, and at the same time be considered valuable and credible by professionals in the field.

I decided to undertake this course and complete the assignment as an additional opportunity to further develop my coding skills and expand my technical background. I was particularly enthusiastic about gaining familiarity with a new programming language that I can now include in my portfolio. Although I initially relied on external resources such as YouTube tutorials and AI tools to understand commands and basic concepts, since I was starting entirely from scratch and alone, I gradually became more confident and autonomous. Furthermore, thanks to my previous experience with HTML, CSS, Java, and jQuery during the Web and Mobile course, I found it easier to understand the logic of the language simply by reading and experimenting with the code, especially compared to more complex languages such as Python or C++.

One aspect I especially appreciated was the realism of the application development process. From the different stages of prototyping to more professional considerations, such as requesting user consent as in a real-world scenario, the course successfully simulated what developing a real application entails. Even when users could simply be friends or acquaintances, treating them as real users reinforced the importance of ethical, legal, and privacy-related considerations. I also enjoyed exploring new and highly requested tools such as Figma, which allowed me to better understand the design and prototyping phase from a professional perspective.

I was particularly enthusiastic about following the entire development lifecycle of an application from scratch, as this was an approach I had already experienced during the Software Engineering project in another DTM course where, despite some challenges, especially related to using Python extensively due to my economic background, I found myself genuinely interested in the engineering process itself. Courses like this, especially when supported by practical projects, are extremely valuable because they help us understand what working in the real job market might look like.

This is especially relevant for our degree program, where we may not necessarily become highly specialized programmers, but rather professionals or managers who need a solid awareness of technical, legal, and organizational processes. Understanding aspects such as privacy, consent, and ethical design, which are topics often taken for granted, helps us realize how much impact our decisions can have in real-world contexts. For example, discussions around paradigm shifts such as moving from “black-list” to “block-list” or using “parent-child” architecture instead of “master-slave” to address bias and discrimination were particularly thought-provoking (and other dark patterns). These are issues that may seem trivial or invisible at first because we are accustomed to them, yet they reveal how much attention is required to avoid ethical and social pitfalls in professional environments.

Overall, I consider this subject useful and fascinating. While it shares similarities with basic web and mobile interface development, it places a strong emphasis on Human-Computer Interaction principles, which are increasingly relevant today. The focus is no longer solely on building functional applications, but on designing complete, integrated user experiences that extend beyond simple smartphone usage, but also touches voice assistants, gestures, sensors and much more that makes the whole environment integrated and fluid and often intertwines with IoT and Industry 4.0. This course has helped me understand the importance of professionalism, inclusivity, and user-centered design, and how crucial it is to meet the needs of diverse users within the specific domain in which an application operates.